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# LoRA-Era Pipeline: Versions, Advantages & Failures

v5.8 to v6.4 - and the consistency gap vs. the previous pipeline

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# LoRA-Era Pipeline: Version History, Advantages & Failures

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This report covers every pipeline version since we adopted LoRA training (v5.8 to v6.4), what each one added, where it helped, where it failed — and an honest assessment of why character consistency is still not at the level of our previous (reference-conditioned) pipeline.

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## v5.8 — LoRA bootstrap (2026-09-07)

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Per-character trained identity on fal.ai: ~10 curated views generated from the character's reference sheet to strict judge curation to fal fast LoRA training to trigger-token registry (ellatok, biscuittok).

### Advantages

- Identity lives in model weights, not just in attached images — in a clean solo smoke test the likeness was essentially perfect.
- Curation loop proved its worth immediately (it dropped the one view missing Ella's charm bracelet).
- Crash-safe registry; ~\$3 per character.

### Failures

- None visible yet at this stage — the problems appeared when the LoRAs met the real page pipeline (v5.9).

## v5.9 — First A/B: the LoRA path LOSES (2026-09-07)

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Pages 1/4/15 of the Ella book rendered pure text-to-image with stacked LoRAs vs. the reference-conditioned baseline.

**Result: baseline 45 / 85 / 75 — LoRA 45 / 45 / 45.**

### Failures (both proven in isolation)

- **Stacking dilution:** two LoRAs active at once interfere; both identities smear (wrong outfit, wrong hair, broken proportions).
- **Token drowning:** trigger tokens carry almost no weight inside a several-hundred-word page prompt (placement + checklist + hex locks + style).

**Advantage of doing it this way:** the 3-page gate stopped us from paying for a full bad book, and gave us the two failure mechanisms to design against.

## v6.0 — Kontext+LoRA: the recipe that finally won (2026-09-07)

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Reference-sheet conditioning AND trained weights in ONE call (`flux-kontext-lora`), with a short prompt.

**Result: 95 / 85 / 75 vs. baseline 45 / 85 / 75 — ≥ baseline on all three.**

### Advantages

- The sheet anchors what things look like; the weights anchor how they are drawn; the short prompt lets both speak.
- Produced the rules we still use: ONE LoRA per page (the main character); co-stars ride in a stitched reference sheet; scale 1.0; short prompt.

### Failures

- Stacking both LoRAs still collapsed the score (45) even with conditioning — confirming one-LoRA-per-page is a hard rule, not a preference.
- Residual small defects (missing bracelet, a leaked in-art sign).

## v6.1 — Recipe wired into the real pipeline (2026-09-07)

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`generate_v3`'s LoRA path became the kontext+LoRA recipe: main character selected by largest layout box, stitched co-star sheets, clean fallback to the reference path.

### Advantages

- 3-page validation: 85 / 95 / 95 vs. baseline 45 / 85 / 75.
- No character-specific hard-coding; mixed casts safe (untrained co-stars are in the conditioning image).

### Failures

- Render variance is real: the same page 15 scored 45 on one roll and 95 on a re-roll (bracelet dropped, busy crowd). Single renders prove nothing.
- The edge-energy "empty band" metric passed frames whose caption area was visibly full of characters (flat cel-shading has few edges).

## v6.2 — Client feedback round on the first full book (2026-09-07)

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Client saw the full Ella book and flagged: tone drifting between pages, the bracelet flipping wrists or vanishing, and no negative space for text.

### Advantages (fixes that came out of it)

- Fixed seed + same-palette prompt sentence for tone stability.
- Compact signature-items line in the prompt ("charm bracelet, always on left wrist"); "each named character exactly ONCE" after a page rendered two Ellas.

- Vision occupancy veto behind the edge-energy metric.
- Discovery that the pipeline was fighting its own art plan: pages that explicitly ask for collage/montage were being "fixed" by the anti-panel rules. Author-intended panel pages now honored.

### Failures (what the client actually experienced)

- **Tone consistency broke** across the book — every page sampled a fresh seed, and nothing anchored palette/lighting between pages.
- **Accessory consistency broke** — the LoRA does not reliably carry ~20-pixel items, and wrist-side flips are a known FLUX mirroring limit.
- **Negative space was not being produced** where the client needed it.

## v6.4 — Bilbo end-to-end (2026-09-08)

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First full second-book run (19-page PDF). Green/blue cap contrast held on every sampled page — the historical duo-collapse defect did not reappear.

### Advantages

- The whole flow (parse to refs to LoRA training to gate to book) now runs hands-off on a new manuscript, ~4× faster page rendering (thread pool).
- Sheet QA loop, phantom-item judge filter, must-appear cast line — all generic hardening that book #3 inherits.

### Failures found (and what they say about the approach)

- Raw renders still dropped or swapped the dogs' caps; one page rendered with ZERO dogs until an explicit cast list was added to the prompt.
- Reference sheets came out wrong three different ways before a QA loop was added (missing caps, wrong species from style-text contamination, wrong character counts).
- The curation judge hallucinated a collar that exists in no spec and no sheet, rejecting 17/20 training views.
- Residuals in the delivered book: a fire-breathing mascot, jersey lettering, coat-fluffiness variance between the two dogs.

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## The honest comparison: consistency vs. the previous pipeline

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The previous pipeline (reference-conditioned generation, and before that the composite engine) delivered **more stable identity out of the raw render**. The 2026-08-26 full Bilbo run on the

composite engine shipped with the D1 defect class essentially gone at render time. The LoRA-era pipeline, by contrast, **depends on the correction loop to converge**:

1. **Raw-render identity is less reliable.** Caps drop, swap between the two dogs, or change color; small accessories (bracelet, hair ties) vanish; in the worst case named characters are omitted entirely. The previous path conditioned every page directly on the sheets with a full lock text, and these defect classes were rarer at the first render.
2. **Tone/style drifts more.** Kontext inherits style from the conditioning image and its own priors; pages in the same book came out in noticeably different finishes until a fixed seed and palette-lock sentence were added — and those mitigate, not eliminate, the drift.
3. **More moving parts = more failure surfaces.** Training-view curation, trigger tokens, LoRA scales, stitched conditioning, prompt length budget — each was a real source of defects this week. The old path had one seam (sheet + prompt to render).
4. **The wins are conditional.** Kontext+LoRA beat baseline convincingly on solo close-ups (95 vs 45 on Ella p1) — weights genuinely help where the camera is close and the character is alone. On multi-character pages the scores tied or the correction loop did the heavy lifting, which the baseline needed less of.

**Bottom line:** kontext+LoRA raised the ceiling (best-case likeness is better than the old pipeline ever produced) but lowered the floor (worst-case raw renders are worse, and consistency across a full book leans on correction). As of v6.4 the end-to-end system converges to a deliverable book, but page-for-page raw consistency has not yet matched the previous reference-conditioned pipeline. The v6\_b feedback round should target exactly that gap.